MACHINE LEARNING PROJECT REPORT

1. Loan Approval Prediction Using Machine Learning

2. House Price Prediction Using Machine Learning

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BATCH	**FEBRUARY -2026**
PROJECT LEVEL	**BEGINNER - INTERMEDIATE**

PROJECT 1 :HOUSE PRICE PREDICTION USING MACHINE LEARNING

ABSTRACT

This project presents a machine learning-based approach for predicting house prices using regression techniques. The dataset contains multiple housing-related features such as crime rate, number of rooms, property tax rate, nitric oxide concentration, and percentage of lower-income population. Comprehensive data preprocessing was performed, including handling missing values, data cleaning, feature selection, train-test splitting, and feature scaling using StandardScaler.

Exploratory Data Analysis (EDA) was conducted using correlation heatmaps to understand relationships between independent variables and the target variable (MEDV). Two regression models—Linear Regression and Random Forest Regressor—were implemented and evaluated using performance metrics such as R^2 score, Mean Absolute Error (MAE), Mean Squared Error (MSE), and Root Mean Squared Error (RMSE).

The Random Forest model achieved superior performance compared to Linear Regression, demonstrating strong predictive capability and good generalization with minimal overfitting. The results indicate that ensemble learning techniques effectively capture complex non-linear relationships in housing data, making them suitable for real-world price prediction applications.

1. INTRODUCTION

House price prediction is a regression problem that estimates the value of residential properties based on multiple influencing factors. Accurate property valuation plays an important role in the real estate industry, financial institutions, and investment planning.

Machine learning algorithms enable automated pattern recognition and predictive modeling from historical data. In this project, regression-based machine learning techniques are applied to develop a reliable house price prediction model.

2. PROBLEM STATEMENT

The objective of this project is to develop a supervised machine learning model that predicts the median value of houses (MEDV) based on various housing-related features.

Problem Type: Supervised Learning – Regression

Target Variable: MEDV (Median value of owner-occupied homes)

3. DATASET DESCRIPTION

The dataset consists of 13 independent features and 1 target variable.

Key features include:

1. CRIM – Crime rate
2. ZN – Residential land proportion
3. INDUS – Non-retail business acres
4. NOX – Nitric oxide concentration
5. RM – Average number of rooms
6. AGE – Age of property
7. DIS – Distance to employment centers
8. TAX – Property tax rate
9. PTRATIO – Pupil-teacher ratio
10. LSTAT – Percentage of lower income population
11. MEDV – Target variable (House price)

These features help analyze how different socio-economic and environmental factors influence housing prices.

4. DATA PREPROCESSING

The following preprocessing steps were applied:

4.1 Handling Missing Values

Missing values were handled using dropna() to ensure clean data.

4.2 Data Cleaning

Non-numeric values were cleaned using regular expressions and converted into numeric format.

4.3 Feature Selection

The target variable was separated from independent variables.

4.4 Train-Test Split

The dataset was split into:

80% Training Data

20% Testing Data

4.5 Feature Scaling

StandardScaler was applied to normalize feature values:

fit_transform() on training data

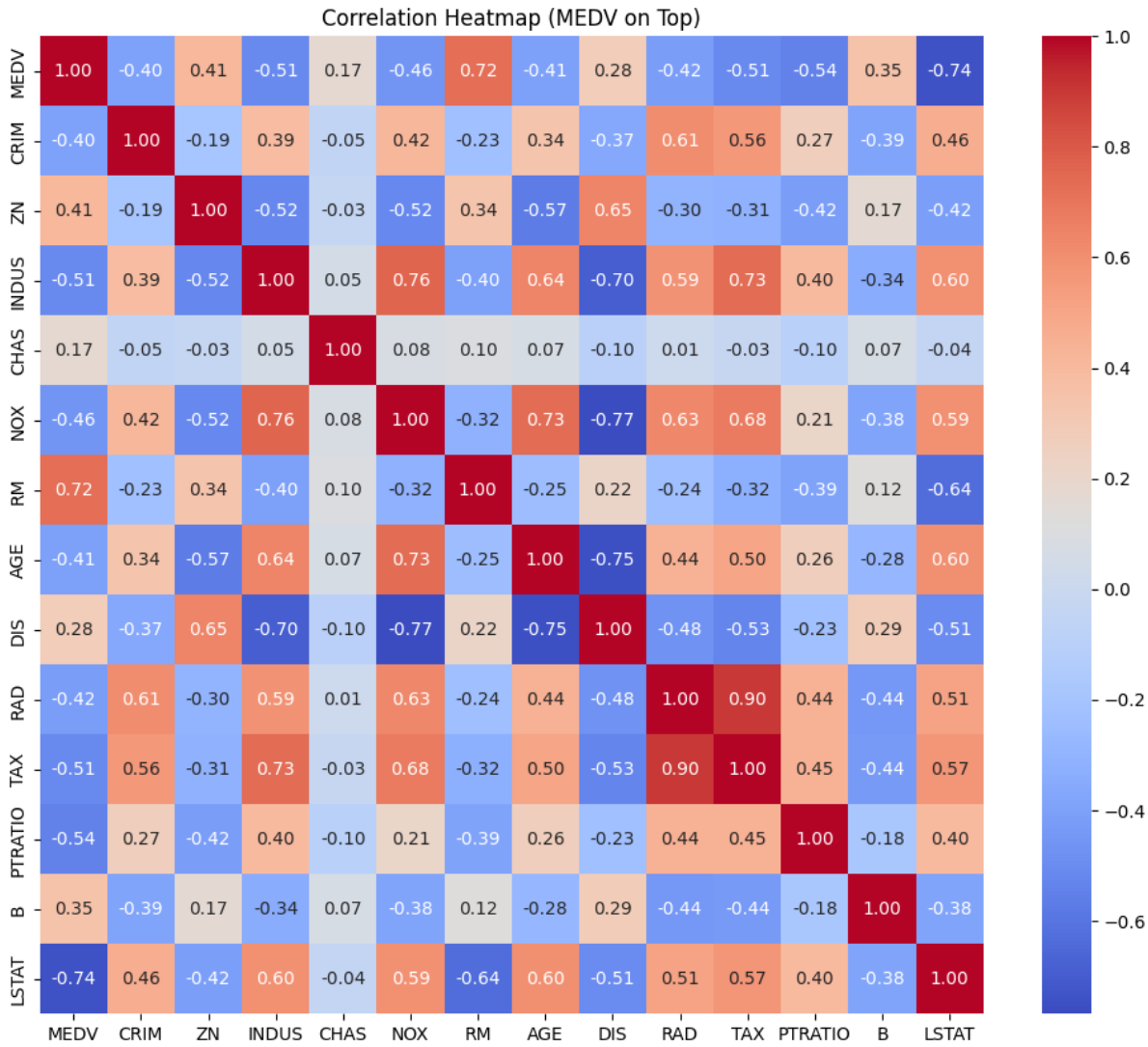
transform() on testing data

5. EXPLORATORY DATA ANALYSIS

A correlation heatmap was generated to examine relationships between features.

Key observations:

- RM shows strong positive correlation with MEDV.
- LSTAT shows strong negative correlation with MEDV.
- Several features moderately influence housing prices.



6. MODEL DEVELOPMENT

Two regression models were implemented:

6.1 Linear Regression

Linear Regression was used as a baseline model to capture linear relationships between features and the target variable.

6.2 Random Forest Regressor

Random Forest is an ensemble learning algorithm that constructs multiple decision trees and combines their predictions to improve accuracy and reduce overfitting.

7. MODEL EVALUATION

The models were evaluated using:

- R^2 Score
- Mean Absolute Error (MAE)
- Mean Squared Error (MSE)
- Root Mean Squared Error (RMSE)

Linear Regression Results:

- Training R^2: 0.74
- Testing R^2: 0.67

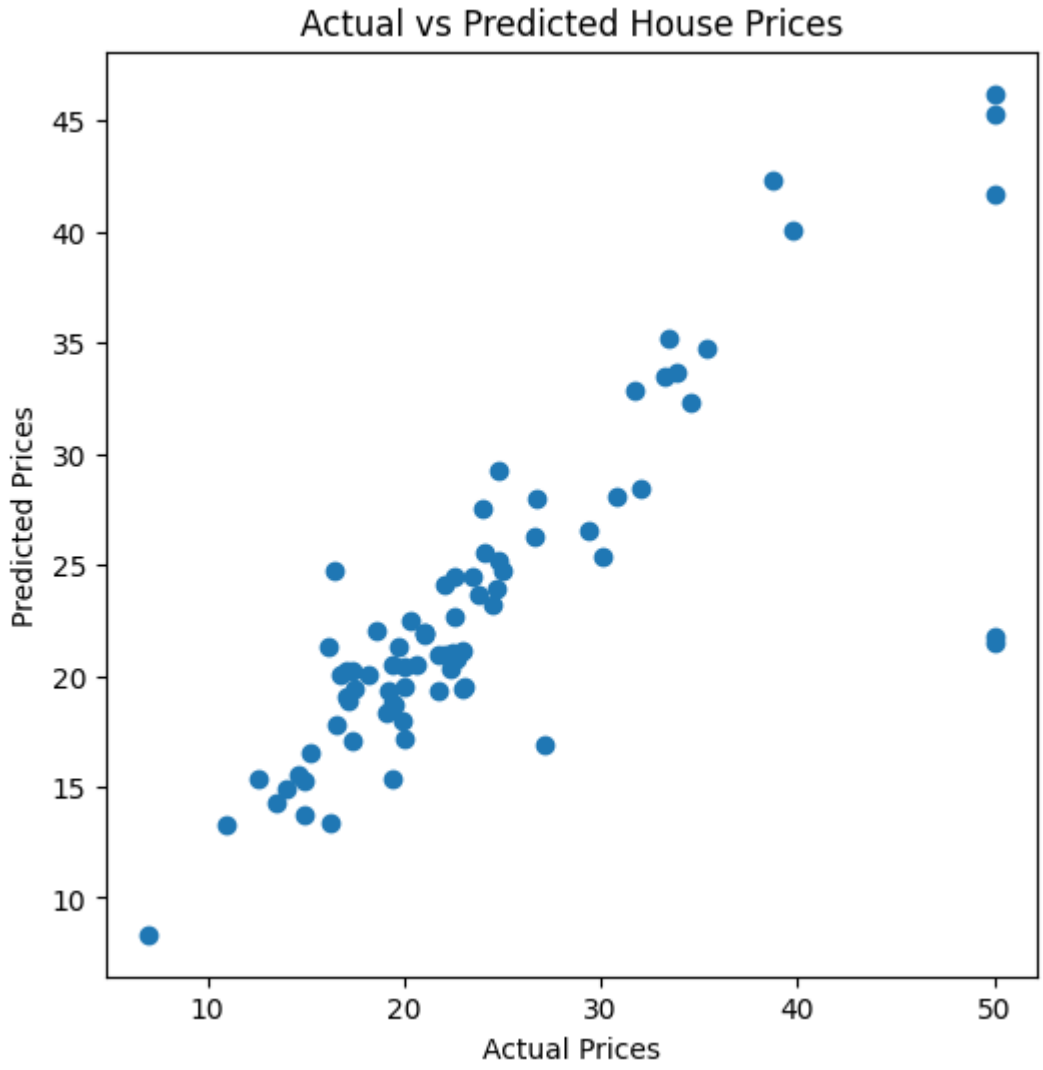
Random Forest Results:

- Training R^2: 0.97
- Testing R^2: 0.85

The Random Forest model outperformed Linear Regression and demonstrated better predictive performance with acceptable overfitting.

7.1 Actual vs Predicted Visualization

A comparison between actual house prices and predicted house prices was plotted to visually evaluate model performance. The closer the points lie to the diagonal line, the better the prediction accuracy of the model.

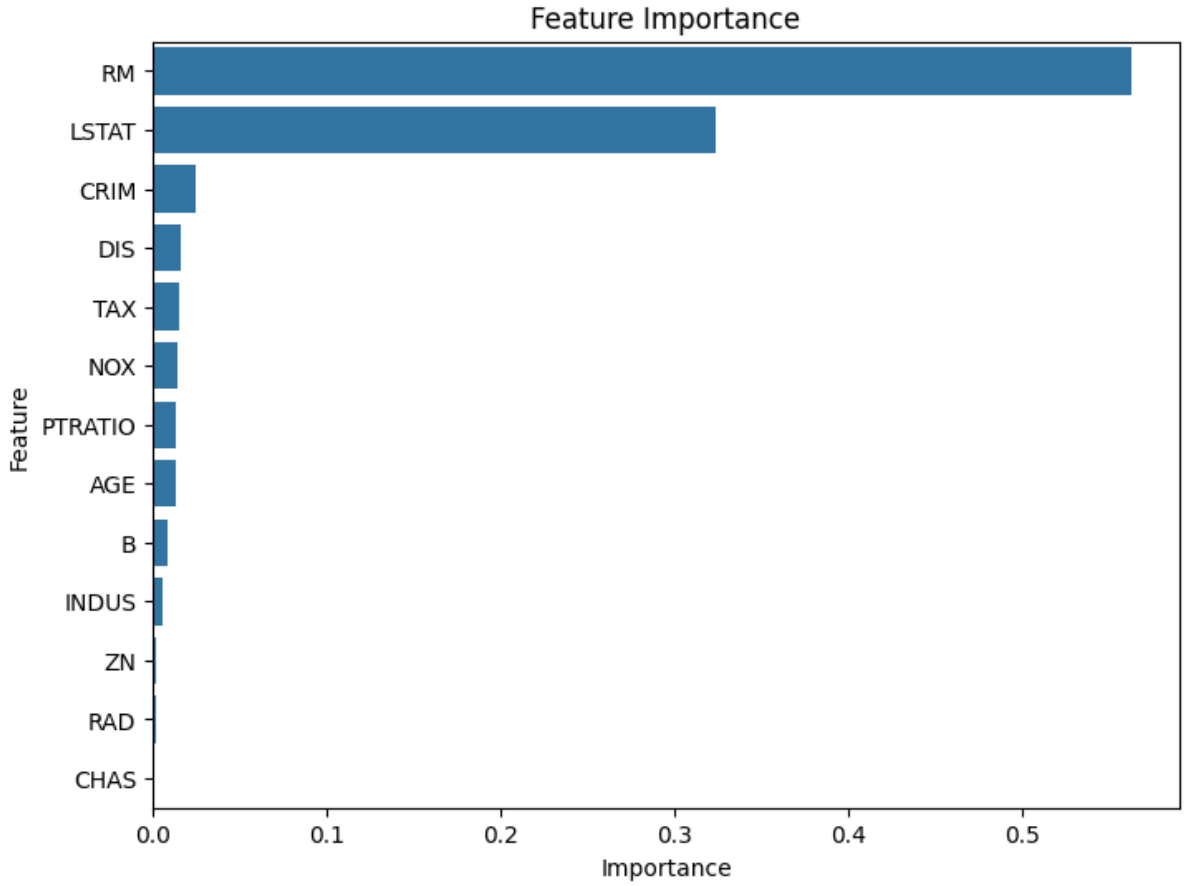


8. FEATURE IMPORTANCE

Feature importance analysis revealed that:

- RM (Number of Rooms)
- LSTAT (Lower Income Percentage)
- PTRATIO

are major contributors influencing house price prediction.



9. CONCLUSION

This project successfully developed a house price prediction system using regression techniques. Random Forest Regressor achieved higher accuracy compared to Linear Regression. The model demonstrates strong generalization ability and effectively captures complex feature relationships.

10. FUTURE IMPROVEMENTS

- Hyperparameter tuning
- Cross-validation
- Implementationof advanced models such as XGBoost
- Model deployment using Flask or Streamlit
- Increasing dataset size

11. REFERENCES

- Scikit-learn Documentation
- Pandas Documentation
- NumPy Documentation
- Matplotlib & Seaborn Documentation

PROJECT 2: Loan Approval Prediction Using Machine Learning

Abstract

Loan approval is one of the most critical decision-making processes in the banking and financial sector. Financial institutions must evaluate multiple applicant attributes such as income level, employment status, credit history, and demographic information before granting a loan. Manual evaluation of such applications can be time-consuming, inconsistent, and prone to human bias. Therefore, machine learning techniques can be effectively used to automate and optimize the loan approval process.

This project focuses on developing a supervised machine learning model to predict whether a loan application will be approved or rejected based on historical applicant data. The dataset was preprocessed through handling missing values, encoding categorical variables, and applying feature scaling to ensure optimal model performance. Three classification algorithms — Logistic Regression, Support Vector Machine (SVM), and Random Forest — were implemented and compared.

Experimental results show that Logistic Regression and SVM achieved an accuracy of 82.11%, while Random Forest achieved 81.30%. The comparable performance of linear models suggests that the dataset exhibits near-linear separability, largely influenced by strong predictive features such as credit history. Based on performance, interpretability, and computational efficiency, Logistic Regression was selected as the final model.

The developed model demonstrates how machine learning can assist financial institutions in improving efficiency, consistency, and objectivity in loan approval decisions.

1.Introduction

In the modern financial ecosystem, loan approval plays a vital role in banking operations. Banks and financial institutions receive thousands of loan applications and must assess each applicant's eligibility based on financial stability, repayment capacity, and creditworthiness. Traditional manual evaluation methods rely heavily on human judgment and predefined rules, which may introduce subjectivity and inefficiencies.

With the advancement of data-driven technologies, machine learning has emerged as a powerful tool for automating decision-making systems. Classification algorithms can analyze historical data patterns and learn relationships between applicant features and loan approval outcomes. These models can then predict future loan approval decisions with significant accuracy.

The primary objective of this project is to design and implement a machine learning-based loan approval prediction system. The system uses applicant demographic and financial attributes such as income, education, employment status, loan amount, loan term, and credit history to classify loan status as Approved or Not Approved.

The workflow of this project includes:

- Data cleaning and preprocessing
- Exploratory data analysis (EDA)
- Feature engineering and encoding
- Model training and evaluation
- Performance comparison of multiple algorithms

Three machine learning models were selected for implementation: Logistic Regression, Support Vector Machine (SVM), and Random Forest Classifier. These models were chosen to compare linear and nonlinear classification techniques.

The final outcome of this project demonstrates that properly preprocessed financial data can effectively be used to build reliable predictive systems that assist financial institutions in making faster and more consistent loan approval decisions.

3.Dataset Description

The dataset contains applicant information:

- Gender
- Married
- Dependents
- Education
- Self_Employed
- ApplicantIncome
- CoapplicantIncome
- LoanAmount
- Loan_Amount_Term
- Credit_History
- Property_Area
- Loan_Status (Target Variable)

Loan_ID was removed as it does not contribute to prediction.

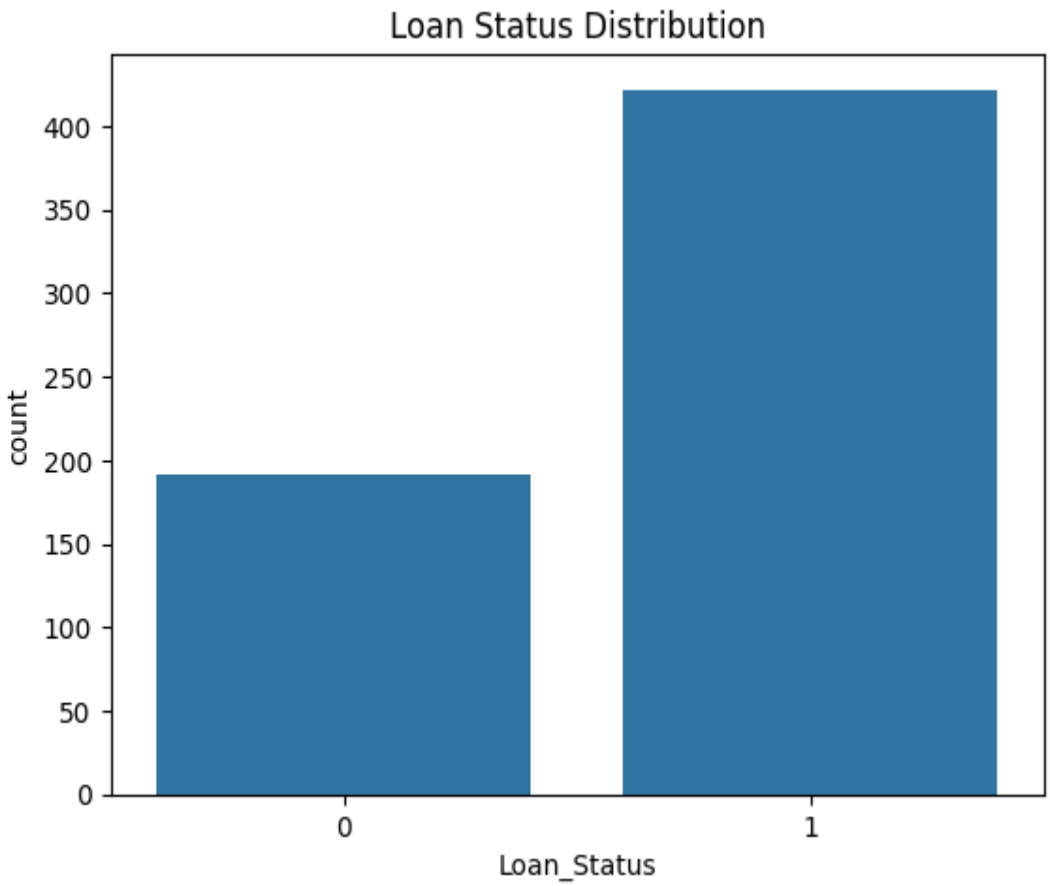
4.Exploratory Data Analysis (EDA)

4.1 Loan Status Distribution

The count plot shows that approved loans are higher than rejected loans.

Interpretation:

- The dataset is slightly imbalanced.
- Most applicants in the dataset were approved.
- Model must handle imbalance carefully.



4.2 Correlation Heatmap

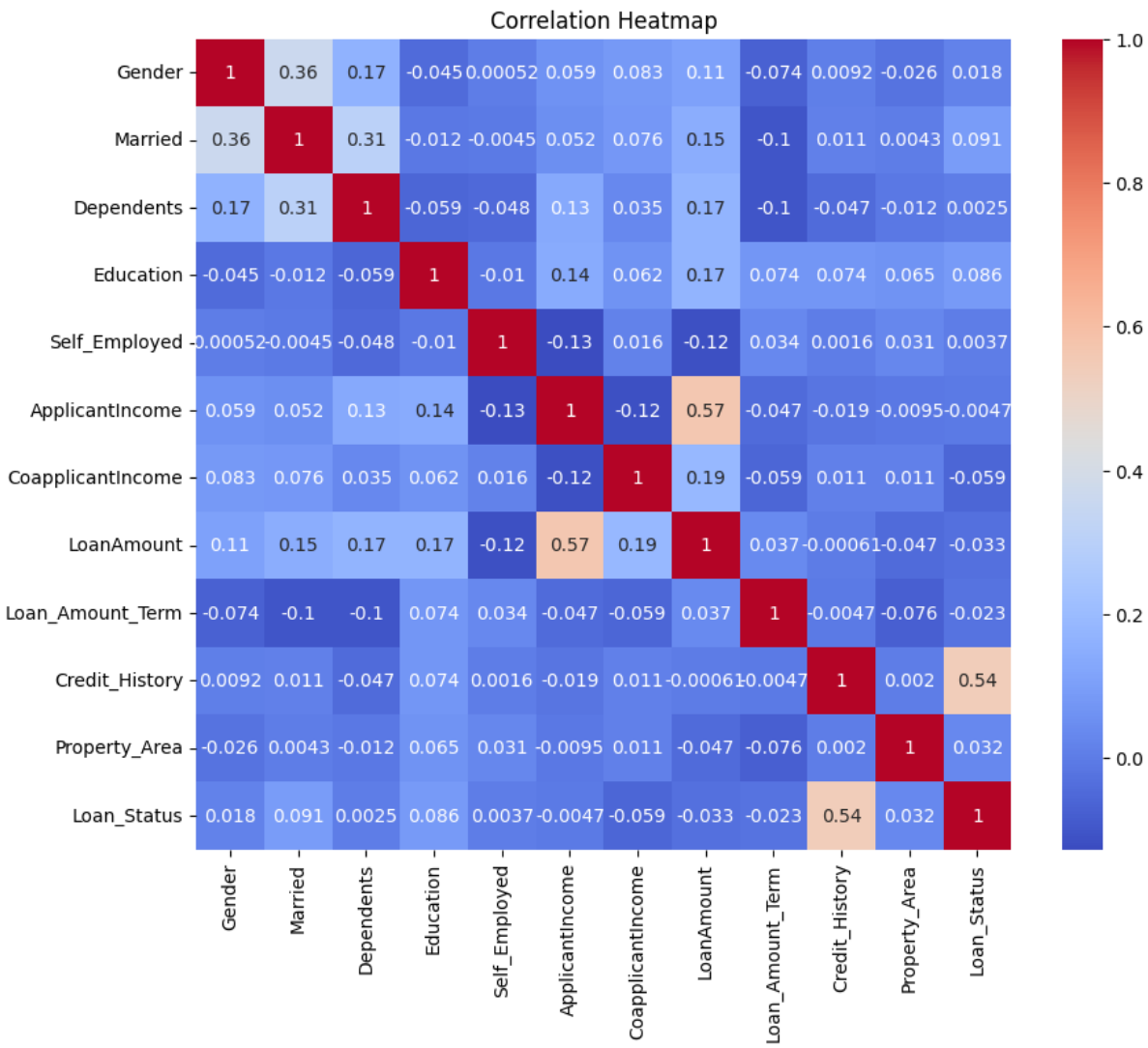
The heatmap shows correlation between features.

Key Observations:

- Credit_History has strong positive correlation with Loan_Status.
- Income features show moderate correlation.
- Most other features have weak correlation.

Interpretation:

Credit_History is the most important predictor of loan approval. This explains why linear models perform well.



5.Data Preprocessing

The following preprocessing steps were applied:

5.1Missing Value Handling

- Mode imputation for categorical variables.
- Median imputation for LoanAmount.

5.2 Encoding

- Converted categorical features into numerical format.
- Yes/No → 1/0
- Graduate → 1
- Not Graduate → 0

5.3 Feature Scaling

- StandardScaler applied.
- Important for SVM and Logistic Regression.

5.4 Train-Test Split

- 80% training
- 20% testing
- Stratified sampling used to maintain class distribution.

6.Model Training

Three classification models were implemented:

1. Logistic Regression
2. Support Vector Machine (Linear Kernel)
3. Random Forest Classifier

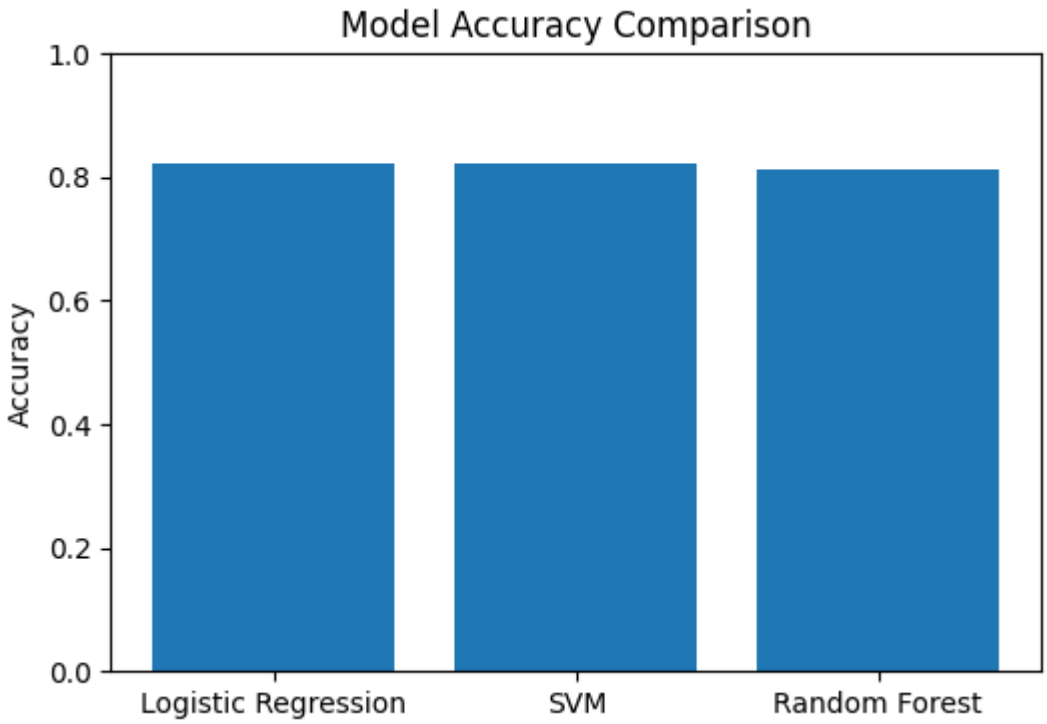
7.Model Evaluation

7.1 Accuracy Comparison

- Logistic Regression: 82.11%
- SVM: 82.11%
- Random Forest: 81.30%

Interpretation:

Logistic Regression and SVM performed identically, indicating the dataset is nearly linearly separable. Random Forest showed slightly lower accuracy.

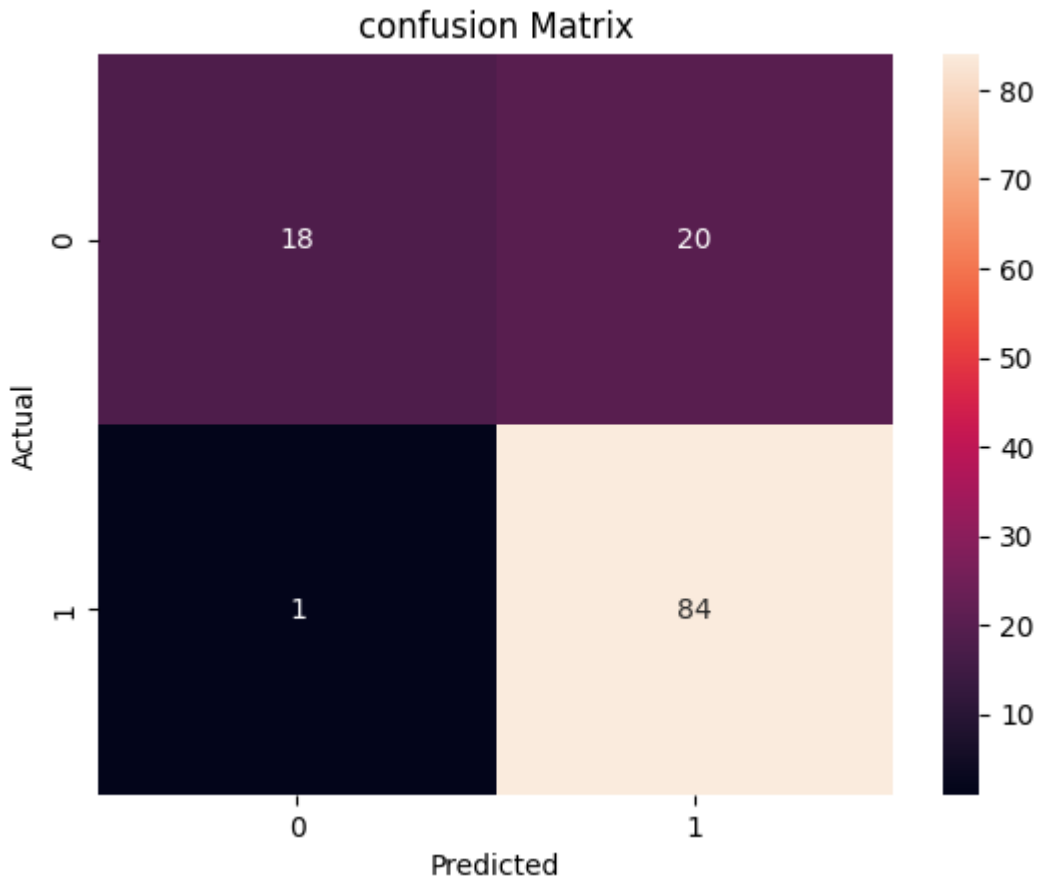


7.2 Confusion Matrix Analysis

The confusion matrix shows:

- High True Positives (correct approvals)
- Moderate False Negatives
- Fewer False Positives

Interpretation:
The model predicts loan approvals effectively but may occasionally reject eligible applicants.



7.3 Classification Report

Precision, Recall, and F1-score were evaluated.

Observations:

- Class 1 (Approved loans) has high precision.
- Recall is slightly lower, indicating some approvals are missed.
- Overall balanced performance.

```
              precision    recall  f1-score   support

           0       0.94      0.42      0.58        38
           1       0.79      0.99      0.88        85

    accuracy                           0.81       123
   macro avg       0.87      0.70      0.73       123
weighted avg       0.84      0.81      0.79       123
```

8. Model Comparison Interpretation

Logistic Regression and SVM performed equally well due to:

- Strong predictive feature (Credit_History)
- Linear separability of dataset
- Limited nonlinear interactions

Random Forest did not significantly improve performance, indicating that the dataset does not require complex nonlinear modeling.

9 Final Model Selection

Final Selected Model: Logistic Regression

Reason:

- Highest accuracy (equal best)
- Simple and interpretable
- Fast training
- Lower computational complexity

10.Conclusion

The project successfully developed a loan approval prediction system using machine learning techniques. After preprocessing and model comparison, Logistic Regression was selected as the final model with 82.11% accuracy.

The results demonstrate that:

- Credit history plays a crucial role in loan approval.
- Linear models perform effectively on this dataset.
- Complex models do not significantly improve performance.
- This system can assist financial institutions in automating loan screening processes.

11. Future Improvements

- Hyperparameter tuning using GridSearchCV
- Cross-validation for robust evaluation
- Feature importance visualization
- Handling class imbalance using SMOTE
- Deployment using Streamlit or Flask
- Integration with real-time financial systems